P. SENTHIL VELAN

AI/ML Engineer | Computer Vision | Generative AI | Full-Stack & IoT

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Portfolio: senz-portfolio.vercel.app

SUMMARY

Final-year B.Tech Artificial Intelligence and Data Science student with 7.74/10 CGPA and hands-on experience building AI/ML, computer vision, Generative AI, cybersecurity, IoT, and full-stack applications. Experienced with Python, Java, TensorFlow/Keras, PyTorch, OpenCV, YOLOv8, LangChain, RAG, Flask, Django, FastAPI, SQL/NoSQL, Docker, ESP32, and cloud deployment. Builds end-to-end systems spanning model development, REST APIs, databases, dashboards, authentication, and real-time monitoring.

EDUCATION

Qualification	Institution	Duration	Score
B.Tech – Artificial Intelligence and Data Science	CARE College of Engineering, Trichy, India	2023 – 2027	CGPA: 7.74/10
Higher Secondary Education (12th)	St. James Matriculation Higher Secondary School, Trichy, India	2021 – 2023	68.67%

CORE TECHNICAL SKILLS

- **Programming:** Python, Java, JavaScript, C++, SQL
- **AI / ML:** Machine Learning, Deep Learning, Computer Vision, NLP, CNN, Transfer Learning, YOLOv8, TensorFlow/Keras, PyTorch, OpenCV, Scikit-Learn
- **Generative AI:** LLMs, Prompt Engineering, RAG, LangChain, IBM Granite, Gemini, Groq, OpenRouter, Ollama
- **Backend:** Flask, Django, FastAPI, REST APIs
- **Data / Cloud:** MySQL, PostgreSQL, MongoDB Atlas, Firebase/Firestore, Pandas, NumPy, Power BI, AWS, GCP, Azure, Supabase
- **DevOps / IoT:** Git, Docker, Gunicorn, Linux, Selenium, Figma, ESP32

EXPERIENCE

Web Development Intern — Jun 2025 – Jul 2025

Aerialroot Infotech

- Applied web development, programming, and UI/UX fundamentals to practical application tasks.

IoT / Project Development Intern — 15 Jun 2026 – 15 Aug 2026

CARE AICTE IDEA Lab

- Completed an internship focused on the FIREGUARD IoT System, involving project development, experimentation, prototyping, and practical problem-solving.

Cloud Computing Intern — Sep 2026 – Oct 2026

CodeAlpha

- Completed a virtual internship program focused on cloud computing concepts and technologies.
- Gained exposure to cloud computing fundamentals through structured, project-based learning.
- Strengthened understanding of cloud infrastructure and practical applications of cloud technology.

SELECTED PROJECTS

TwinFed-OT – Privacy-Preserving Federated Cybersecurity Platform — *Cybersecurity / Federated Learning*

- Designed a cybersecurity research platform combining Federated Learning, Graph Neural Networks, Digital Twin simulation, SBOMs, and vulnerability-package analysis for collaborative OT threat detection and safer patch validation.
- Kept sensitive operational data distributed while using Digital Twin simulation to evaluate patch effects before deployment.

ISL Recognition System – YOLOv8 Computer Vision — *Computer Vision / Object Detection*

- Trained a custom 36-class Indian Sign Language object-detection model using YOLOv8; documented training run scanned 41,076 training images and 240 validation images.
- Optimized training with 416x416 inputs, batch size 32, 35 epochs, and YOLOv8s GPU training for real-time gesture detection and fine-tuning.

FireGuard IoT – Real-Time Fire Monitoring and Alert System — *IoT / ESP32 / Flask*

- Architected an ESP32-to-Flask IoT pipeline with serial JSON ingestion and 3 monitoring API routes for latest readings, history, and alerts.
- Engineered a browser dashboard for live sensor visualization, alert states, and historical event tracking.

AI Assistant – Full-Stack Conversational AI (Gemini + Flask + MongoDB Atlas) — *Generative AI / Full-Stack*

- Engineered a 2-layer conversational AI architecture combining a deterministic FAQ engine with Google Gemini for open-ended responses.
- Integrated MongoDB Atlas conversation history, Web Speech API voice input, REST health checks, Docker containerization, and Google Cloud Run deployment.

RESPIRONE – ESP32 Wearable IoT + On-Device ML

- Team-built an ESP32 wearable combining sensor acquisition, live dashboarding, vibration/LED alerts, and on-device machine-learning concepts for real-time monitoring.

ADDITIONAL PROJECTS

EcoMetal LCA – Full-Stack AI-Powered Life Cycle Assessment

- Engineered a React + FastAPI + PostgreSQL platform using scikit-learn, SHAP, SQLAlchemy, Alembic, JWT security, and AI API integration for emissions prediction and sustainability analytics.

SkinAI – CNN-Based Medical Image Classification

- Developed a TensorFlow/Keras pipeline using MobileNetV2 transfer learning across 16 disease classes, 224x224 preprocessing, batch size 32, and up to 30 epochs.
- Implemented image augmentation, early stopping, model checkpointing, classification reports, and confusion-matrix evaluation.

EON Multilingual Chatbot – Django / NLP / Translation

- Engineered a Django chatbot platform with custom models for students, faculty, departments, and subjects; translation layer maps 17 languages including English, Tamil, Hindi, Telugu, Malayalam, Spanish, French, German, Chinese, and Japanese.

Uzhavan Market – Full-Stack Agricultural Price and Alert Platform

- Engineered a Flask/MySQL application with 2 authentication workflows: farmer JWT and administrator session.
- Implemented crop and mandi search, subscriptions, price alerts, ratings, parameterized SQL queries, password hashing, and responsive UI.

SkillLearn – Full-Stack AI-Powered Personalized Learning Platform

- Engineered a Flask/MySQL learning platform with JWT, bcrypt, Groq API integration, personalized learning workflows, challenges, performance tracking, XP, streaks, and badges.

DataGuard – Data Redundancy Removal System

- Built a Flask + Cloud Firestore application with 3 primary collections: records, dedupe_keys, and review_queue.
- Implemented contact normalization, SHA-256-derived uniqueness keys, atomic transactions, exact duplicate blocking, CSV import, and manual review for possible same-name matches.

CodeAlpha Bus – Cloud-Based Bus Pass System

- Engineered a Flask application using PostgreSQL/Cloud SQL, SQLAlchemy, QR verification, administrator authentication, automatic pass expiry, health checks, Docker/Gunicorn deployment, and pytest.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **IBM SkillsBuild:** Getting Started with AI; LLM Basics; Prompt Engineering; AI Document Retrieval; RAG with LangChain
- **AWS:** Machine Learning for Natural Language Processing
- **Cognitive Class:** Data Analysis with Python
- **HackerRank:** SQL Basic; Python Basic; Java Basic; Problem Solving Basic; Problem Solving Intermediate
- **MongoDB:** MongoDB Basics
- **Forage:** Deloitte Australia – Data Analytics Job Simulation (Jul 2026); Siemens Mobility – Operations Industrial Engineer Job Simulation (Jul 2026)
- **Other:** Coursera – Computer Vision: YOLOv8; Infosys Springboard – Artificial Intelligence, Cyber Security; Udemy – Machine Learning; AI Tools Workshop (Oct 2025)